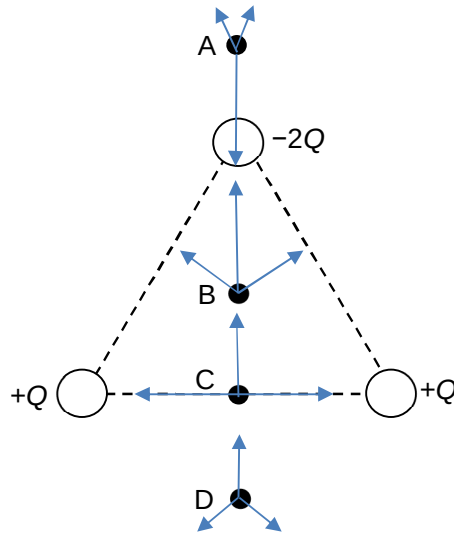


Q17 The forces due to the charges at different positions are shown.

2024/JPJC/PHYSICS/9749



$$F \propto \frac{q_1 q_2}{r^2}$$

At A, the force due to the negative charge $-2Q$ is much larger than the vector sum of forces due to the 2 positive charges $+Q$. Hence, there is a net downward force.

At B and C, the direction of resultant force on test charge is upwards.

Answer: D

$$Q18 \quad I = \frac{1}{2} n q v_d A \quad (a) \quad n(b) \quad nq \quad (c) \quad nqV$$